

ETHAN MAH

Software Developer — Mobile, Automotive & Web

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SUMMARY

Software developer with 3+ years of production experience across Android mobile development, automotive infotainment, and full-stack web projects. Shipped production features to a 500K+ user consumer app at Jiffy on Demand and built Android Automotive OS infotainment software on pre-release vehicles at General Motors. Experienced in delivering end-to-end features across mobile, web, and automotive platforms, and in leveraging AI-assisted development tools to accelerate delivery.

TECHNICAL SKILLS

Languages	Kotlin, Java, C++, Python, JavaScript, TypeScript, SQL
Android	Android Automotive OS (AAOS), Jetpack Compose, MVVM, Clean Architecture, Coroutines, Flow, Hilt, Room, Firebase, XML Layouts, JNI
Web & Frontend	React, TypeScript, CSS, REST APIs, Firebase (Firestore, Auth, Hosting), Google Apps Script
Automotive	AOSP, Android Auto, CarPlay-adjacent infotainment stack, GM VIP Platform, CAN Bus Signals, ARXML, Hardware-in-the-Loop Testing
AI & Tooling	Claude Code, LLM-assisted development, Fastlane, CI/CD, Firebase App Distribution, Google Play Console
Testing	JUnit, Espresso, Test-Driven Development (TDD)
Tools & Methods	Git, JIRA, Gradle, Android Studio, Agile/Scrum, Code Reviews, Technical Documentation

EXPERIENCE

Jiffy on Demand — Junior Android Developer

Toronto, ON | Jul 2023 – Present

Kotlin · Jetpack Compose · MVVM · Clean Architecture · Hilt · Room · Firebase · Agile/Scrum · 500K+ user app

- Designed, built, and shipped end-to-end production features in Kotlin across multiple core user flows including New Customer Promotional Boosters, Schedule & Save, Job Details Revamp, and Credit Card Fraud Detection.
- Contributed to an ongoing XML-to-Jetpack Compose migration, refactoring legacy UI components within MVVM and Clean Architecture using Coroutines, Hilt, and Room while maintaining stability across active user flows.
- Leveraged Claude Code and AI-assisted development tools to accelerate feature delivery, automate repetitive tasks, and improve code quality — reducing time spent on boilerplate and increasing throughput.
- Implemented Fastlane to automate deployment pipelines to Firebase App Distribution and Google Play Console — an independent initiative that reduced manual release overhead for the Android team.
- Drove cross-platform feature alignment as the Android point of contact during planning and delivery, collaborating across a 10-person tech team (iOS, Android, Web) and participating in regular code reviews to reinforce architecture consistency.

General Motors — Software Developer, Infotainment

Markham, ON | May–Aug 2022, Jan–Apr 2023

Co-op · Java · Android Automotive OS (AAOS) · AOSP · GM VIP Platform · CAN · ARXML · JNI · Emulator Development

- Implemented end-to-end software for vehicle hard-key telephony controls, enabling drivers to answer and end calls via console and multi-function controller buttons — coordinating across vehicle signal handling, Android Automotive's telephony stack, and HMI teams.
- Built and shipped mute and volume control services for GM's Android Automotive infotainment system in Java, validated against pre-release vehicle builds and hardware configurations.
- Enabled location services for the passenger-side infotainment display in GM's multi-display architecture, supporting location-aware applications independent of the driver display.
- Developed infotainment features on GM's VIP platform, working with CAN bus signals and ARXML communication descriptions to integrate Android Automotive software with the vehicle's software-defined architecture.
- Built and maintained internal setup and configuration tooling for GM's next-generation Android Automotive emulator during a critical hardware-to-emulator transition — adopted org-wide across multiple infotainment teams; served as cross-team subject-matter resource for emulator onboarding.

PROJECTS

Nexus Sports League — Web App & Automation

React · TypeScript · Firebase · Google Apps Script

- Designed and built a full-stack web application for a community sports league, featuring league standings, schedules, and player management using React and TypeScript on the frontend with Firebase as the backend.
- Developed a Google Apps Script automation pipeline to sync league data between Google Sheets and the web app, eliminating manual data entry and reducing administrative overhead for league coordinators.
- Owned the full product lifecycle from UI/UX design through deployment and ongoing maintenance — serving as the sole developer for an active multi-team league.

EDUCATION

Queen's University

Kingston, ON | 2020 – 2025

Bachelor of Computing, Software Design Specialization · Minor in Business · Certificate in Entrepreneurship & Innovation